

Input(1): 今天上午 9 点半, <u>复旦投毒案</u> 将在上海 <u>二中院公开审理</u>。被害学生 <u>黄洋</u> 的亲属已从四川抵达上海, <u>其父</u> 称待刑事部分结束后, 再提民事赔偿, <u>黄洋</u> 92 岁的奶奶依然不知情。今年 4 月, 在复旦 <u>上海医学院</u> 读研究生的 <u>黄洋</u> 疑遭室友 <u>林森浩</u> 投毒, 不幸身亡。新民网 Today 9:30, the Fudan poisoning case will be will on public trial at the Shanghai Second Intermediate Court. The relatives of the murdered student Huang Yang has arrived at Shanghai from Sichuan. His father said that they will start the lawsuit for civil compensation after the criminal section. HuangYang 92-year-old grandmother is still unaware of his death. In April, a graduate student at Fudan University Shanghai Medical College, Huang Yang is allegedly poisoned and killed by his roommate Lin Senhao. Reported by Xinmin Golden: 林森浩投毒案今日开审 92岁奶奶尚不知情 the case of Lin Senhao poisoning is on trial today, his 92-year-old grandmother is still unaware of this RNN context: 复旦投毒案: 黄洋疑遭室友投毒凶手已从四川飞往上海, 父亲命案另有4人被通知家属不治? CopyNet: <u>复旦投毒案</u> 今在沪上公开审理 the Fudan poisoning case is on public trial today in Shanghai
Input(2): 华谊兄弟 (<u>300022</u>) 在昨日收盘后发布公告称, 公司拟以自有资金 <u>3.978亿元</u> 收购浙江 <u>永乐影视</u> 股份有限公司若干股东持有的 <u>永乐影视</u> 51% 的股权。对于 <u>此项</u> 收购, 华谊兄弟 <u>董秘</u> <u>胡明</u> 昨日表示: “和 <u>永乐</u> 影视的合并是对华谊兄弟 <u>电视剧业务</u> 的一个 <u>加强</u>。” Huayi Brothers (300022) announced that the company intends to buy with its own fund 397.8 million 51% of Zhejiang Yongle Film LTD's stake owned by a number of shareholders of Yongle Film LTD. For this acquisition, the secretary of the board, Hu Ming, said yesterday: "the merging with Yongle Film is to strengthen Huayi Brothers on TV business". Golden: 华谊兄弟拟收购永乐影视 51% 股权 Huayi Brothers intends to acquire 51% stake of Zhejiang Yongle Film RNN context: 华谊兄弟收购永乐影视51%股权: 与永乐影视合并为“和唐”影视合并的“UNK”和“UNK”的区别? CopyNet: <u>华谊兄弟</u> 拟 <u>3.978 亿</u> 收购 <u>永乐影视</u> <u>董秘</u> 称 <u>加强</u> <u>电视剧业务</u> Huayi Brothers is intended to 397.8 million acquisition of Yongle Film secretaries called to strengthen the TV business
Input(3): 工厂, 大门紧锁, 约 <u>20 名工人</u> 散坐在 <u>树荫下</u>。“我们就是普通工人, 在这里等工资。”其中一人说道。7月4日上午, 记者抵达深圳龙华区清湖路上的深圳 <u>愿景</u> <u>光电子</u> 有限公司。正如传言一般, <u>愿景</u> <u>光电子</u> <u>倒闭</u> 了, 大股东 <u>邢毅</u> 不知所踪。 The door of factory is locked. About 20 workers are scattered to sit under the shade. "We are ordinary workers, waiting for our salary" one of them said. In the morning of July 4th, reporters arrived at Yuanjing Photoelectron Corporation located at Qinghu Road, Longhua District, Shenzhen. Just as the rumor, Yuanjing Photoelectron Corporation is closed down and the big shareholder Xing Yi is missing. Golden: 深圳亿元级LED企业倒闭烈日下工人苦等老板 Hundred-million CNY worth LED enterprise is closed down and workers wait for the boss under the scorching sun RNN context: 深圳“<UNK>”: 深圳<UNK><UNK>, <UNK>, <UNK>, <UNK> CopyNet: <u>愿景光电子</u> <u>倒闭</u> <u>20 名工人</u> 散坐在 <u>树荫下</u> Yuanjing Photoelectron Corporation is closed down, 20 workers are scattered to sit under the shade
Input(4): 截至 2012 年 10 月底, 全国累计报告艾滋病 <u>病毒感染者</u> 和病人 <u>492191</u> 例。<u>卫生部</u> 称, <u>性传播</u> 已成为 <u>艾滋病</u> 的 <u>主要传播途径</u>。至 2011 年 9 月, <u>艾滋病感染者</u> 和 <u>病人</u> 数累计报告 <u>数</u> 挂在前 6 位的省份依次为云南、广西、河南、四川、新疆和广东, 占全国的 <u>75.8%</u>。 At the end of October 2012, the national total of reported HIV infected people and AIDS patients is 492,191 cases. The Health Ministry said sexual transmission has become the main route of transmission of AIDS. To September 2011, the six provinces with the most reported HIV infected people and AIDS patients were Yunnan, Guangxi, Henan, Sichuan, Xinjiang and Guangdong, accounting for 75.8% of the country. Golden: 卫生部: 性传播成艾滋病主要传播途径 Ministry of Health: Sexually transmission became the main route of transmission of AIDS RNN context: 全国累计报告艾滋病患者和病人<UNK>例艾滋病患者占全国<UNK>%, 性传播成艾滋病高发人群? CopyNet: <u>卫生部</u>: <u>性传播</u> 已成为 <u>艾滋病</u> <u>主要传播途径</u> Ministry of Health: Sexually transmission has become the main route of transmission of AIDS
Input(5): <u>中国反垄断</u> 调查风暴 <u>继续席卷汽车行业</u>, 继德国 <u>车企</u> 奥迪和美国 <u>车企</u> <u>克莱斯勒</u> “<u>沦陷</u>”之后, 又有 <u>12 家日本汽车企业</u> <u>卷入漩涡</u>。记者从业内人士获悉, 丰田旗下的 <u>雷克萨斯</u> 近期 <u>曾被发改委约谈</u>。 Chinese antitrust investigation continues to sweep the automotive industry. After Germany Audi car and the US Chrysler "fell", there are 12 Japanese car companies involved in the whirlpool. Reporters learned from the insiders that Toyota's Lexus has been asked to report to the Development and Reform Commission recently. Golden: 发改委公布汽车反垄断进程: 丰田雷克萨斯近期被约谈 the investigation by Development and Reform Commission: Toyota's Lexus has been asked to report RNN context: 丰田雷克萨斯遭发改委约谈: 曾被约谈丰田旗下的雷克萨斯遭发改委约谈负责人被约谈 CopyNet: <u>中国反垄断</u> <u>继续席卷汽车行业</u> <u>12 家日本汽车企业</u> <u>被发改委约谈</u> Chinese antitrust investigation continues to sweep the automotive industry. 12 Japanese car companies are asked to report to the Development and Reform Commission
Input(6): <u>镁离子</u> 电池相比锂电池能量密度提升了近一倍, 这意味着使用了 <u>镁</u> 电池的电动车, <u>续航里程</u> 也将 <u>有质的提升</u>。但目前由于 <u>电解液</u> 等 <u>技术壁垒</u>, <u>要大规模量产并取代锂电池</u> 还为时过早。 The energy density of Magnesium ion batteries almost doubles that of lithium battery, which means that for the electric vehicles using of magnesium batteries will last longer even at pure electric power. But currently due to the technical barriers to the electrolyte, it is still too early for the mass production of it and replacing lithium batteries.. Golden: 锂电池或将被淘汰 能量密度更高的镁电池亦大势所趋 Lithium batteries will be phased out, magnesium battery with energy density higher will be the future trend RNN context: <UNK>, <UNK>, <UNK>, <UNK>, <UNK>, <UNK>, <UNK>, <UNK>, <UNK>, <UNK>, <UNK>, <UNK> 电池了 CopyNet: <u>镁离子</u> 电池问世: <u>大规模量产</u> <u>取代锂电池</u> Magnesium ion battery is developed: mass production of it will replace lithium batteries
Input(7): 1. 掌握技巧 <u>融会贯通</u>; 2. 学会融资; 3. 懂法律; 4. 保持自信; 5. 测试+尝试; 6. 了解客户的需求; 7. 预测+衡量+确保; 8. 做好与各种小 <u>bug</u> 做斗争的心态; 9. 发现机遇保持 <u>创业</u> 激情。 1. master the skills; 2 Learn to finance; 3. understand the law; 4. Be confident; 5. test+ trial; 6. understand the need of customers; 7 forecast + measure + ensure; 8. mentally prepared to fight all kinds of small bugs; 9 discover opportunities and keep the passion of start-up. Golden: 初次创业者必知的 10 个技巧 The 10 tips for the first time start-ups RNN context: 6 个方法让你创业的 6 个<UNK>与<UNK>, 你怎么看懂你的创业故事吗? (6 家) CopyNet: <u>创业</u> 成功的 <u>9 个</u> 技巧 The 9 tips for success in start-up
Input(8): 9 月 3 日, 总部位于 <u>日内瓦</u> 的世界经济论坛发布了《<u>2014 - 2015 全球竞争力报告</u>》, <u>瑞士</u> 连续六年位居榜首, 成为全球最具竞争力的国家, 新加坡和美国 <u>分列第二位</u> 和 <u>第三位</u>。<u>中国</u> 排名第 <u>28 位</u>, 在 <u>金砖</u> 国家中排名最高。 On September 3, the Geneva based World Economic Forum released "The Global Competitiveness Report 2014-2015". Switzerland topped the list for six consecutive years, becoming the world's most competitive country. Singapore and the United States are in the second and third place respectively. China is in the 28th place, ranking highest among the BRIC countries. Golden: 全球竞争力排行榜中国居 28 位居金砖国家首位 The Global competitiveness ranking list, China is in the 28th place, the highest among BRIC countries. RNN context: 2014-2015 年全球竞争力报告: 瑞士连续 6 年居榜首中国居 28 位(首/3——访榜首)中国排名第 28 位 CopyNet: <u>2014 - 2015 年</u> <u>全球竞争力报告</u>: <u>瑞士</u> 居首 <u>中国</u> 第 <u>28</u> 2014 - 2015 Global Competitiveness Report: Switzerland topped and China the 28th

Figure 4: Examples of COPYNET on LCSTS compared with RNN context. Word segmentation is applied on the input, where OOV words are underlined. The highlighted words (with different colors) are those words with copy-mode probability higher than the generate-mode. We also provide literal English translation for the document, the golden, and COPYNET, while omitting that for RNN context since the language is broken.

one. One possible explanation is that a word-based model, even with a much larger vocabulary (50,000 words in Hu et al. (2015)), still has a large proportion of OOVs due to the large number of entity names in the summary data and the mistakes in word segmentation. COPYNET, with its ability to handle the OOV words with the copying mechanism, performs however slightly better with the word-based variant.

5.2.1 Case Study

As shown in Figure 4, we make the following interesting observations about the summary from COPYNET: 1) most words are from copy-mode, but the summary is usually still fluent; 2) COPYNET tends to cover consecutive words in the original document, but it often puts together segments far away from each other, indicating a sophisticated coordination of content-based addressing and location-based addressing; 3) COPYNET handles OOV words really well: it can generate acceptable summary for document with many OOVs, and even the summary itself often contains many OOV words. In contrast, the canonical RNN-based approaches often fail in such cases.

It is quite intriguing that COPYNET can often find important parts of the document, a behavior with the characteristics of extractive summarization, while it often generate words to “connect” those words, showing its aspect of abstractive summarization.

5.3 Single-turn Dialogue

In this experiment we follow the work on neural dialogue model proposed in (Shang et al., 2015; Vinyals and Le, 2015; Sordoni et al., 2015), and test COPYNET on single-turn dialogue. Basically, the neural model learns to generate a response to user’s input, from the given (input, response) pairs as training instances.

Dataset: We build a simple dialogue dataset based on the following three instructions:

1. Dialogue instances are collected from Baidu Tieba³ with some coverage of conversations of real life e.g., greeting and sports, etc.
2. Patterns with slots like

$$hi, my\ name\ is\ x \rightarrow hi, x$$
are mined from the set, with possibly multiple responding patterns to one input.

³<http://tieba.baidu.com>

3. Similar with the synthetic dataset, we enlarge the dataset by filling the slots with suitable subsequence (e.g. name entities, dates, etc.)

To make the dataset close to the real conversations, we also maintain a certain proportion of instances with the response that 1) do not contain entities or 2) contain entities not in the input.

Experimental Setting: We create two datasets: DS-I and DS-II with slot filling on 173 collected patterns. The main difference between the two datasets is that the filled substrings for training and testing in DS-II have no overlaps, while in DS-I they are sampled from the same pool. For each dataset we use 6,500 instances for training and 1,500 for testing. We compare COPYNET with canonical RNNSearch, both character-based, with the same model configuration in Section 5.1.

Models	DS-I (%)		DS-II (%)	
	Top1	Top10	Top1	Top10
RNNSearch	44.1	57.7	13.5	15.9
COPYNET	61.2	71.0	50.5	64.8

Table 4: The decoding accuracy on the two testing sets. Decoding is admitted success only when the answer is found exactly in the Top-K outputs.

We compare COPYNET and RNNSearch on DS-I and DS-II in terms of top-1 and top-10 accuracy (shown in Table 4), estimating respectively the chance of the top-1 or one of top-10 (from beam search) matching the golden. Since there are often many good responses to an input, top-10 accuracy appears to be closer to the real world setting.

As shown in Table 4, COPYNET significantly outperforms RNNsearch, especially on DS-II. It suggests that introducing the copying mechanism helps the dialogue system master the patterns in dialogue and correctly identify the correct parts of input, often proper nouns, to replicate in the response. Since the filled substrings have no overlaps in DS-II, the performance of RNNSearch drops significantly as it cannot handle words unseen in training data. In contrast, the performance of COPYNET only drops slightly as it has learned to fill the slots with the copying mechanism and relies less on the representation of the words.

5.3.1 Case Study

As indicated by the examples in Figure 5, COPYNET accurately replicates the critical segments from the input with the copy-mode, and generates

Input(1): 我出生在 <u>上海</u> 。 /I was born in Shanghai	Golden: 真巧, 我也来自上海, 老乡啊! /What a coincidence, I also come from Shanghai, we are fellow townsman!
	✗ RNNSearch: 真巧, 我也来自海南, 老乡啊! /What a coincidence, I also come from Hainan, we are fellow townsman!
	○ CopyNet: 真巧, 我也来自 <u>上海</u> , 老乡啊! /What a coincidence, I also come from Shanghai, we are fellow townsman!
Input(2): 今天去爬 <u>黄山</u> 吗? /Shall we climbing Huangshan?	Golden: 去爬泰山吧, 黄山没意思! /Climb Taishan. Huangshan is boring!
	✗ RNNSearch: 去爬长白山吧, 衡山没意思! /Climb Changbaishan, Hengshan is boring!
	○ CopyNet: 去爬 <u>嵩山</u> 吧, <u>黄山</u> 没意思! /Climb Songshan, Huangshan is boring!
Input(3): 你喜 <u>欢</u> 喝 <u>雪碧</u> 还是喜 <u>欢</u> <u>可</u> <u>乐</u> ? /Would you like to drink Sprite or Coke?	Golden: 都不喜欢, 谢谢! /Not at all, thanks!
	✗ RNNSearch: 我喜欢喝蜂蜜! /I like to drink honey!
	○ CopyNet: 我喜 <u>欢</u> 喝 <u>可</u> <u>乐</u> ! /I like to drink coke!
Input(4): 我大 <u>青</u> <u>岛</u> 四季宜人 /Our great Qingdao has a good weather!	Golden: 壮我大青岛~~~ /Our Qingdao is great~~~
	✗ RNNSearch: 壮我大哈尔滨~~~ /Our Harbin is great~~~
	○ CopyNet: 壮我大 <u>青</u> <u>岛</u> ~~~ /Our Qingdao is great~~~
Input(5): <u>女</u> <u>儿</u> 会游 <u>蝶</u> <u>泳</u> ? /Daughter can swim butterfly?	Golden: 我连游泳都不会T^T /I cannot swim at all T^T
	○ RNNSearch: 我连游泳都不会T^T /I cannot swim at all T^T
	○ CopyNet: 我连游 <u>泳</u> 都不会T^T /I cannot swim at all T^T
Input(6): <u>苏</u> <u>有</u> <u>朋</u> 是谁? /Who is Su You Peng?	Golden: 苏有朋是一个男明星。 /Su You Peng is a male star.
	✗ RNNSearch: 吴亦凡是一个男明星。 /Wu Yifan is a male star.
	✗ CopyNet: <u>苏</u> <u>有</u> 是一个男明星。 /Su You is a male star

Figure 5: Examples from the testing set of DS-II shown as the input text and golden, with the outputs of RNNSearch and CopyNet. Words in red rectangles are unseen in the training set. The highlighted words (with different colors) are those words with copy-mode probability higher than the generate-mode. Green circles (meaning correct) and red cross (meaning incorrect) are given based on human judgment on whether the response is appropriate.

the rest of the answers smoothly by the generate-mode. Note that in (2) and (3), the decoding sequence is not exactly the same with the standard one, yet still correct regarding to their meanings. In contrast, although RNNSearch usually generates answers in the right formats, it fails to catch the critical entities in all three cases because of the difficulty brought by the unseen words.

6 Related Work

Our work is partially inspired by the recent work of Pointer Networks (Vinyals et al., 2015a), in which a pointer mechanism (quite similar with the proposed copying mechanism) is used to predict the output sequence directly from the input. In addition to the difference with ours in application, (Vinyals et al., 2015a) cannot predict outside of the set of input sequence, while COPYNET can naturally combine generating and copying.

COPYNET is also related to the effort to solve the OOV problem in neural machine translation. Luong et al. (2015) introduced a heuristics to post-process the translated sentence using annotations on the source sentence. In contrast COPYNET addresses the OOV problem in a more systemic way with an end-to-end model. However, as COPYNET copies the exact source words as the output, it cannot be directly applied to machine translation. However, such copying mechanism can be naturally extended to any types of references except for the input sequence, which will help in applications with heterogeneous source and target se-

quences such as machine translation.

The copying mechanism can also be viewed as carrying information over to the next stage without any nonlinear transformation. Similar ideas are proposed for training very deep neural networks in (Srivastava et al., 2015; He et al., 2015) for classification tasks, where shortcuts are built between layers for the direct carrying of information.

Recently, we noticed some parallel efforts towards modeling mechanisms similar to or related to copying. Cheng and Lapata (2016) devised a neural summarization model with the ability to extract words/sentences from the source. Gulcehre et al. (2016) proposed a pointing method to handle the OOV words for summarization and MT. In contrast, COPYNET is more general, and not limited to a specific task or OOV words. Moreover, the softmaxCOPYNET is more flexible than gating in the related work in handling the mixture of two modes, due to its ability to adequately model the content of copied segment.

7 Conclusion and Future Work

We proposed COPYNET to incorporate copying into the sequence-to-sequence learning framework. For future work, we will extend this idea to the task where the source and target are in heterogeneous types, for example, machine translation.

Acknowledgments

This work is supported in part by the China National 973 Project 2014CB340301.